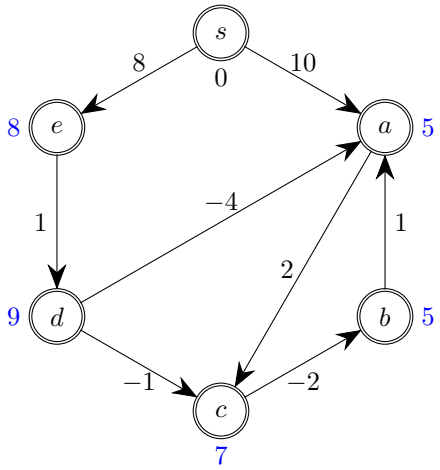
Bellman Ford Examples

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Graph

https://www.youtube.com/watch?v=obWXjtg0L64



Relaxation Queue

Step	Queue	s	a	b	c	d	e
0	`[s]`	0	∞	∞	∞	∞	∞
1	`[a, e]`	0	10	∞	∞	∞	8
2	`[e, c]`	0	10	∞	12	∞	8
3	`[c, d]`	0	10	∞	12	9	8
4	`[d, b]`	0	10	10	12	9	8
5	`[b, a, c]`	0	5	10	8	9	8
6	`[a, c]`	0	5	10	8	9	8
7	`[c]`	0	5	10	7	9	8
8	`[b]`	0	5	5	7	9	8
9	`[]`	0	5	5	7	9	8

Dynamic Programming

Recursion:

$$\texttt{OPT}(v,i) = \min \begin{cases} \texttt{OPT}(v,i-1) \\ \min_{w \in N(v)} \left[\texttt{OPT}(w,i-1) + d(w,v)\right] \end{cases}$$